

E016

Risk Management and Information Technology

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A clinic that sees 200 patients a day keeps records on a file server in a locked closet. The server runs software with a known, unpatched flaw. Criminal groups are known to target clinics with ransomware. Which item is the vulnerability?

- A. The criminal groups targeting clinics.
- B. The locked closet.
- C. The possible loss of patient records.
- D. The unpatched software flaw.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A customer database, migrated to new hardware in 2024, is valued at \$200,000. A data-loss event would destroy 25% of its value, and such an event is expected once every two years. What is the annualized loss expectancy?

- A. \$50,000
- B. \$25,000
- C. \$12,500
- D. \$100,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A risk has an ALE of \$30,000 and an annual rate of occurrence of 0.2. What is its single loss expectancy?

- A. \$150,000
- B. \$6,000
- C. It cannot be found without the asset value.
- D. \$30,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

During the annual audit, held in the company's new conference center, a risk is found with an ALE of \$2,000. The only available control would cost \$20,000 a year. The risk falls within the board's stated risk appetite. What is the most appropriate response?

- A. Buy the \$20,000 control.
- B. Ignore it without recording anything.
- C. Eliminate every risk, whatever the cost.
- D. Formally accept the risk and document the decision.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A data center built in 2019 and powered partly by solar panels installs a system that alerts staff whenever unusual login activity occurs. What type of control is this?

- A. Preventive.
- B. Detective.
- C. Corrective.
- D. Preventive, since all security tools stop attacks.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Which of these is a corrective control?

- A. Reviewing security logs daily.
- B. Requiring multifactor authentication.
- C. An acceptable-use policy.
- D. Restoring systems from backups after an incident.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

Customer records live in a cloud region in Virginia and are maintained by a database administrator with 12 years of experience. Who is accountable for classifying the data and deciding who may access it?

- A. The database administrator.
- B. The data owner — the business leader responsible for that data.
- C. The cloud provider.
- D. The end users who read the records.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A risk has an inherent ALE of \$500,000. Controls recommended by a well-known consulting firm reduce it by 70%. The board's risk appetite for this category is \$100,000. What is the residual risk, and what follows?

- A. \$150,000; it exceeds appetite, so more treatment is needed.
- B. \$350,000; it exceeds appetite.
- C. \$150,000; acceptable because controls were applied.
- D. \$0, since controls eliminate risk.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

On a Friday afternoon, the security team confirms malware on several marketing-department workstations, and it appears to be spreading. Following the incident response lifecycle, what should the team do next?

- A. Wait until Monday, since it started on a Friday.
- B. Write the lessons-learned report.
- C. Contain it by isolating the affected workstations from the network.
- D. Immediately reimaging every computer in the company.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

When should an organization hold its lessons-learned review for an incident?

- A. During containment.
- B. In the preparation phase, before any incident.
- C. After recovery, as post-incident activity.
- D. Only when an incident causes financial loss.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A risk has an ALE of \$80,000. A proposed control would cut the ALE to \$20,000 and cost \$45,000 a year. The vendor offers a free first month. What is the control's annual value, and is it justified?

- A.** +\$60,000; justified.
- B.** Justified because the first month is free.
- C.** +\$15,000; justified.
- D.** -\$25,000; not justified.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

12. [TRANSFER — reversed stem / different variable]

A risk has an ALE of \$80,000. A control costs \$30,000 a year. What is the highest ALE after the control at which the control still breaks even?

- A.** \$50,000
- B.** \$0, since a control must remove all risk to be worth it.
- C.** \$30,000
- D.** \$110,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A clinic ~~that sees 200 patients a day~~ keeps records on a file server ~~in a locked closet~~. The server runs software with a known, unpatched flaw. Criminal groups are known to target clinics with ransomware. Which item is the vulnerability?

Target word: identify the vulnerability **Decides:** The weakness is the unpatched flaw.

A	Confusion set	They are the threat.
B	Camouflage	A physical control, not the weakness in question.
C	Adjacent category	That is the impact that makes up the risk.
D	Correct	A vulnerability is a weakness that could be exploited.

2. Answer B Concept 2

A customer database, ~~migrated to new hardware in 2024~~, is valued at \$200,000. A data-loss event would destroy 25% of its value, and such an event is expected once every two years. What is the annualized loss expectancy?

Target word: ALE **Decides:** $SLE = \$200,000 \times 0.25 = \$50,000$; $ARO = 0.5$.

A	Dropped element	That is the SLE; it drops the ARO.
B	Correct	$\$50,000 \times 0.5 = \$25,000$.
C	Confusion set	Applies the 25% exposure factor twice.
D	Inverted operation	Divides SLE by the ARO instead of multiplying.

3. Answer A Concept 2 · transfer

A risk has an ALE of \$30,000 and an annual rate of occurrence of 0.2. What is its single loss expectancy?

Target word: solve for SLE **Decides:** $SLE = ALE / ARO$.

A	Correct	$\$30,000 / 0.2 = \$150,000$.
B	Inverted operation	Multiplies ALE by ARO.
C	Confusion set	SLE can be backed out of ALE and ARO directly.
D	Wrong stem number	That is the ALE.

4. Answer D Concept 3

During the annual audit, ~~held in the company's new conference center~~, a risk is found with an ALE of \$2,000. The only available control would cost \$20,000 a year. The risk falls within the board's stated risk appetite. What is the most appropriate response?

Target word: treatment choice **Decides:** Control cost far exceeds expected loss; within appetite.

A	Inverted operation	Spends \$20,000 a year to prevent \$2,000 of expected loss.
B	Dropped element	Acceptance must be a documented, owned decision.
C	Absolute language	Risk management balances cost and benefit.
D	Correct	Acceptance is valid when a risk is within appetite and controls cost more than they save.

5. Answer B Concept 4

A data center ~~built in 2019 and powered partly by solar panels~~ installs a system that alerts staff whenever unusual login activity occurs. What type of control is this?

Target word: control function **Decides:** It identifies events; it does not stop or fix them.

A	Confusion set	Preventive controls stop events before they happen.
B	Correct	Alerting identifies an event as or after it happens.
C	Adjacent category	Corrective controls restore systems after an event.
D	Absolute language	Many tools detect or correct rather than prevent.

6. Answer D Concept 4 · transfer

Which of these is a corrective control?

Target word: pick the corrective control — reverse direction **Decides:** Acts after an event to restore.

A	Confusion set	Detective: it finds events.
B	Inverted operation	Preventive: it acts before an event.
C	Adjacent category	An administrative, mainly preventive control.
D	Correct	It returns systems to normal operation.

7. Answer B Concept 5

Customer records live ~~in a cloud region in Virginia~~ and are maintained by a database administrator ~~with 12 years of experience~~. Who is accountable for classifying the data and deciding who may access it?

Target word: accountable role **Decides:** Classification and access decisions belong to the owner.

A	Confusion set	A custodian who implements controls, not the decision-maker.
B	Correct	Owners are accountable for classification and access.
C	Camouflage	Hosting location does not transfer accountability.
D	Adjacent category	Users follow access rules; they do not set them.

8. Answer A Concept 6

A risk has an inherent ALE of \$500,000. Controls ~~recommended by a well-known consulting firm~~ reduce it by 70%. The board's risk appetite for this category is \$100,000. What is the residual risk, and what follows?

Target word: residual risk vs. appetite **Decides:** 30% of \$500,000 remains.

A	Correct	$\$500,000 \times 0.30 = \$150,000$, above \$100,000.
B	Inverted operation	That is the amount reduced, not what remains.
C	Dropped element	Drops the comparison with appetite.
D	Absolute language	Some risk always remains.

9. Answer C Concept 7

~~On a Friday afternoon,~~ the security team confirms malware on several ~~marketing department~~ workstations, and it appears to be spreading. Following the incident response lifecycle, what should the team do next?

Target word: next IR phase **Decides:** Detection is confirmed; spread is ongoing.

A	Camouflage	Timing does not justify letting it spread.
B	Inverted operation	Post-incident review comes last.
C	Correct	Containment limits damage before eradication and recovery.
D	Adjacent category	Jumps to broad eradication before containing and scoping.

10. Answer C Concept 7 • transfer

When should an organization hold its lessons-learned review for an incident?

Target word: place of lessons learned **Decides:** Post-incident activity follows recovery.

A	Inverted operation	The team is still stopping the damage.
B	Confusion set	Preparation builds capability; there is no incident yet to learn from.
C	Correct	Lessons are drawn once the full incident is understood.
D	Absolute language	Every significant incident offers lessons.

11. Answer C Concept 8

A risk has an ALE of \$80,000. A proposed control would cut the ALE to \$20,000 and cost \$45,000 a year. ~~The vendor offers a free first month.~~ What is the control's annual value, and is it justified?

Target word: net value of the control **Decides:** $\$80,000 - \$20,000 - \$45,000$.

A	Dropped element	Drops the control's annual cost.
B	Camouflage	A promotion does not change the annual comparison.
C	Correct	The \$60,000 reduction exceeds the \$45,000 cost.
D	Confusion set	Uses the residual ALE as the benefit instead of the reduction.

12. Answer A Concept 8 • transfer

A risk has an ALE of \$80,000. A control costs \$30,000 a year. What is the highest ALE after the control at which the control still breaks even?

Target word: solve for ALE after **Decides:** Reduction must at least equal the \$30,000 cost.

A	Correct	$\$80,000 - \$30,000 = \$50,000$.
B	Absolute language	A partial reduction can more than pay for itself.
C	Wrong stem number	The control's cost, not the break-even ALE.
D	Inverted operation	Adds the cost instead of subtracting it.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Threat, vulnerability, and risk	A vulnerability is a weakness; a threat is something that could exploit it; risk is the likelihood and impact of that happening.	Calling the attacker the vulnerability or the possible loss the threat.	A weakness, an adversary, and a possible loss in one scenario.
2	Annualized loss expectancy	$SLE = \text{asset value} \times \text{exposure factor}$; $ALE = SLE \times \text{annual rate of occurrence}$.	Stopping at SLE or dividing by the ARO.	Asset value, a percentage loss, and a frequency per year.
3	Risk treatment options	Avoid, transfer, mitigate, or accept; accepting a documented risk within appetite is valid when controls cost more than they save.	Believing every risk must be reduced regardless of cost.	A small expected loss and an expensive control.
4	Control functions	Preventive controls stop an event; detective controls identify it; corrective controls restore after it.	Calling alerting tools preventive.	Whether the control acts before, during, or after an event.
5	Data governance roles	The data owner (a business leader) is accountable for classification and access decisions; the custodian implements and maintains the controls.	Assigning ownership to IT staff or the cloud provider.	"Who decides who can access this data?"
6	Residual risk and risk appetite	Residual risk is what remains after controls; it must fall within the organization's risk appetite or be treated further.	Assuming controls bring risk to zero.	An inherent risk, a percentage reduction, and an appetite limit.
7	Incident response lifecycle	Preparation; detection and analysis; containment, eradication, and recovery; then post-incident lessons learned.	Jumping to cleanup before containing, or holding lessons learned too early.	"What should the team do next?" during an incident.
8	Control cost-benefit	A control is justified when the ALE reduction exceeds its annual cost: $\text{value} = ALE_{\text{before}} - ALE_{\text{after}} - \text{annual control cost}$.	Ignoring the control's cost or using residual ALE as the benefit.	ALE before, ALE after, and a yearly control price.

Course scope (public description): The Risk Management course approaches an organization's risk needs from a management perspective. This course will enable the student to understand how risk management protects their organization's sensitive data, intellectual property, and digital assets from threats and breaches. This knowledge includes understanding governance structures, developing a mitigation strategy that includes risks, and creating an incident response plan. There are no prerequisites for this course.